



Enhancing the Efficiency and Stability of Inverted Perovskite Solar Cells and Modules through Top Interface Modification with N-type Semiconductors

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Abstract: The interface modification between perovskite and electron transport layer (ETL) plays a crucial role in achieving high performance inverted perovskite photovoltaics (i-PPVs). Herein, non-fullerene acceptors (NFAs), known as Y6-BO and Y7-BO, were utilized to modify the perovskite/ETL interface in i-PPVs. Non-polar solvent-soluble NFAs can effectively passivate surface defects without structural damage of the underlying perovskite films. Additionally, the improved phenyl-C61-butyric acid methyl ester (PCBM) ETL induced by NFAs modification significantly accelerates the electrons extraction. As a result, both Y6-BO and Y7-BO exhibit more effective interface modification effects compared to traditional PI molecules. The power conversion efficiency (PCE) of the inverted perovskite solar cell (i-PSC) modified with Y7-BO reaches 25.82 %. Moreover, the adoption of non-polar solvents and the superior semiconductor properties of Y7-BO molecules also enable perovskite solar modules (i-PSM) with effective areas of 50 cm², 400 cm², and 1160 cm² to achieve record efficiencies of 23.05 %, 22.32 %, and 21.1 % (certified PCE), respectively, making them the best PCE reported in the literature. Importantly, enhanced interface mechanical strength between the perovskite and PCBM layer results in significantly improved environmental and operational stability of the cells. The cells modified with Y7-BO maintained 94.4 % of the initial efficiency after 1522 hours of maximum power point aging.

Introduction

Recently, organic–inorganic hybrid perovskite photovoltaics (PPVs) have attracted much attention as the rapid increasing in their power conversion efficiencies (PCEs), thanks to large absorption coefficient, high charge mobility, long charge diffusion length, etc., of the perovskite material.^[1–3] Among various device configurations, planar inverted PPVs (p-i-n structure, i-PPVs) are currently sitting in the spotlight due to their low processing temperature and potential using in flexible devices.^[4,5] The small-size inverted perovskite solar cells (i-PSCs) have reached outstanding performances

over the past decade,^[6–8] while, the performances of inverted perovskite solar modules (i-PSMs) decreased obviously when the size of active layer increased.^[9] The top interface between the perovskite and electron transport layer (ETL) plays a key role in the performance boosting of i-PSMs, which not only influence the device efficiency, but also affect the device stability.^[10] Hence, interfacial optimization at perovskite/ETL interface is essential to obtain high efficient and stable i-PSM devices.

Phenyl-C61-butyric acid methyl ester (PCBM), a derivative of fullerene, was by far the most commonly used ETLs in i-PPVs.^[11–13] In this type device, the property of perov-

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skite/PCBM interface is mainly affected by the upper surface of perovskite layer and buried surface of PCBM layer. On the one hand, most of defects residing on the perovskite upper surface and grain boundaries (GBs) can cause charge nonradiative recombination, reducing open-circuit voltage (V_{oc}) and fill factor (FF) of devices.^[14] Besides, surface defects are prone to adsorb oxygen and water molecules in the air, leading to the degradation of the perovskite crystals and thus the device stability.^[15] On the other hand, low solubility and viscosity properties of PCBM cause a nonuniform and defect-filled film,^[16] resulting in incomplete defect passivation on the perovskite upper surface, and causing weak interactions between PCBM and perovskite.^[17] In addition, the forming pinholes in the ETL due to the agglomeration of PCBM and interspaces among the spherical structured PCBM molecules may serve as oxygen, water, and ion migration channels, impairing the stability of devices. Therefore, simultaneously engineering the perovskite upper surface and PCBM buried surface at perovskite/PCBM interface with one single material is highly desirable to enhance the efficiency and stability of i-PPVs.

For solvent-based interface modifiers, both the properties of the modifier and effects of the solvent on the underlying perovskite must be considered. Organic ammonium salts (OASs) are the most commonly used interface functional materials,^[18–22] but the use of inappropriate solvent isopropyl alcohol (IPA) can induce perovskite phase decomposition during surface treatment. Non-fullerene acceptors (NFAs),^[23–33] soluble in low-polarity solvents such as chlorobenzene (CB) and immiscible with water, represent superior agents for surface modification of perovskite. They can efficiently passivate defects without altering the structure or properties of the underlying perovskite layer. It remains uncertain whether interface engineering with NFAs is applicable to large-sized perovskite solar modules.

In this work, Y6-BO and Y7-BO NFAs dissolved in CB solvent were applied to optimize the perovskite/PCBM interface of i-PPVs, which is scalable and can be combined with various printing methods for the production, like spin-coating for small-size i-PSCs and slot-die coating for large-area i-PSMs. The NFAs can simultaneously passivate defects on the upper surface of perovskite and optimize the buried surface of PCBM layer, enhancing electron extraction at the top interface for highly efficient i-PPVs. Furthermore, Y7-BO can lead to stronger interfacial interaction between perovskite and PCBM than that of Y6-BO NFA, thereby promoting electron transfer more efficiently in the resultant devices. As a result, the PCE increased from 24.49 % to 25.12 %, and 25.82 % for the control, Y6-BO-, and Y7-BO-treated i-PSCs. Particularly, the i-PSMs with Y7-BO NFAs exhibited excellent PCEs of 23.06 %, 22.32 %, and 21.10 % (certified efficiency) for the 50 cm², 400 cm², and 1160 cm² devices. Moreover, compared to the control devices treated by piperazinium iodide (PI) in IPA solvent, the NFAs-based devices exhibit excellent stability due to the pure defect passivation without disrupting the underlying perovskite layer and improved uniformity of PCBM ETL. Consequently, the interface stabilized i-PSCs and i-PSMs based on

Y7-BO NFAs maintain 90 % of the initial efficiency after 1400 h at a relative humidity (RH) condition of 40 % and 94.4 % of its initial efficiency after operating for 1522 h under maximum power point (MPP) tracking, respectively.

Results and Discussion

The perovskite film with surface treatment of PI (a rationally designed bifunctional molecule that can interact with both positive and negative charged defects^[34]) in IPA solvent was prepared as a reference, the perovskite films treated with Y6-BO and Y7-BO NFAs was prepared by dissolved them in CB solvent which was dripped on the upper surface of perovskite films (Figure 1a). Hereafter, the perovskite (PVK) films with Y6-BO and Y7-BO NFAs are denoted as PVK/Y6-BO and PVK/Y7-BO, respectively. Notably, CB and IPA solvents did not effectively dissolve the PI and NFAs (Figure S1), respectively. Thus, PI in CB and NFAs in IPA cannot be used for surface treatment. The scanning electron microscopy (SEM) images (Figure 1b–d and Figure S2) display that the control perovskite film shows clear grain boundaries (GBs) on the upper surface, while GBs of the NFAs-treated ones become indistinct. The stitched GBs can resist the moisture and oxygen penetrating into the film, which is beneficial to prevent perovskite degradation and improving the stability of the resultant photovoltaic devices. The X-ray diffraction (XRD) patterns in Figure S3 show that those NFA-treated perovskite films exhibit the strongest (001) peak intensity, indicating that the PVK/Y6-BO and PVK/Y7-BO films present higher crystallinity than that of the control one. To certify NFAs existing on the PVK films, we tested the UV/Vis absorption spectrum and X-ray photoelectron spectroscopy (XPS). Both PVK/Y6-BO and PVK/Y7-BO films exhibit light absorption in the near-infrared range after 800 nm, typically belonging to the NFA materials (Figure S4).^[35] The XPS spectra in Figure 1e–g show that PVK/Y6-BO and PVK/Y7-BO films present typical signals of F and organic Cl, confirming the presence of Y6-BO and Y7-BO NFAs on the perovskite upper surface, as further demonstrated by the Time-of-flight secondary ion mass spectrometry (TOF-SIMS, Figure S5).

Y6-BO and Y7-BO with S, N, and O-contained functional groups could react with Pb clusters or undercoordinated Pb²⁺ to form complexes on the top perovskite surface.^[27] To investigate the interaction between NFAs and perovskite, density functional theory (DFT) calculation was adopted. Figure S6 present the optimized surface structure of the perovskites interacted with Y6-BO and Y7-BO. Particularly, the flexible alkyl chains is near-vertical to the core units when stacking on the perovskite surface in the optimal configuration, this behavior was also observed in the aggregate structure of the Y6-BO molecules in organic solar cell.^[35] Figure 2a and 2b display their corresponding charge-density difference (CDD) isosurface distributions. We see that charge depletion regions mostly distributed around the core units of NFAs in both two cases, indicating large amount of electrons are transferred to the perovskite surface, thereby passivating surface defects.^[36] Since the

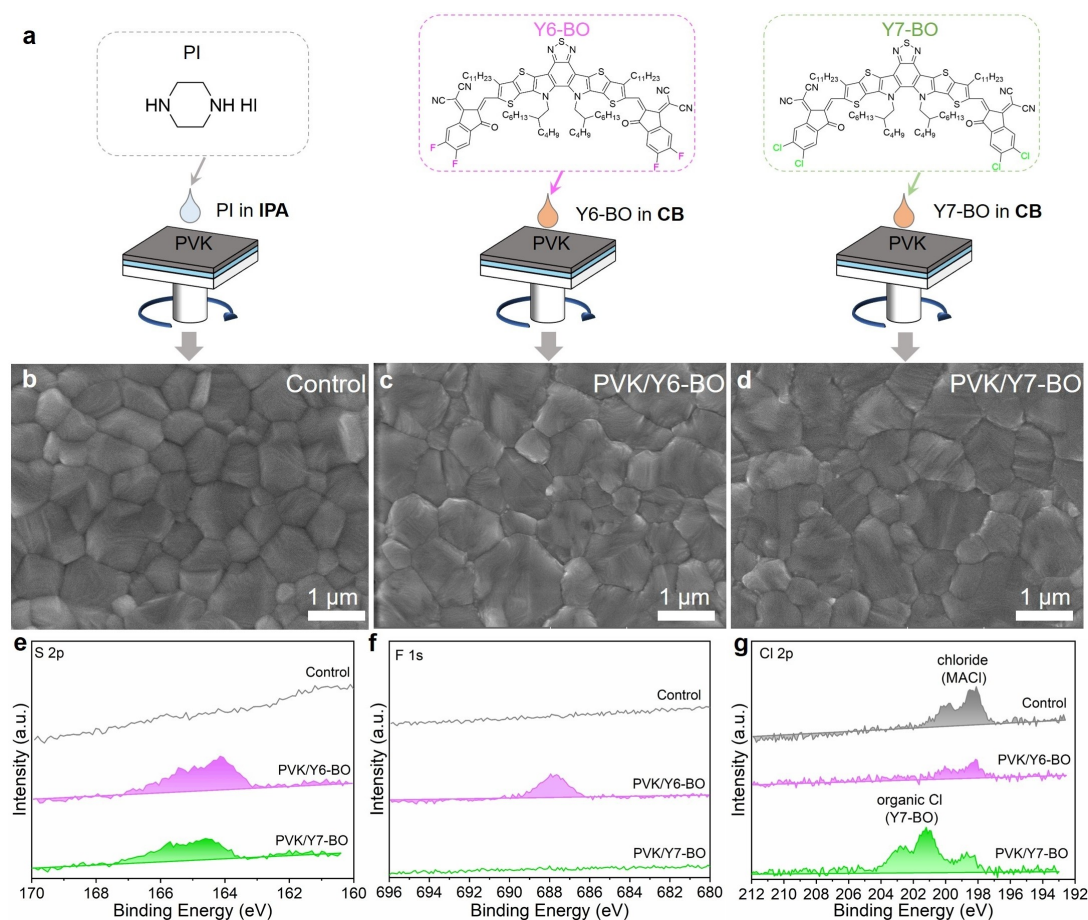


Figure 1. (a) Chemical structure of PI, Y6-BO and Y7-BO, and illustration of the deposition processes of perovskite films. (b-d) SEM images of Control, PVK/Y6-BO and PVK/ Y7-BO films. XPS spectra of (e) S 2p, (f) F 1s and (g) Cl 2p of Control, PVK/Y6-BO and PVK/Y7-BO films.

empty 3d orbitals of Cl atom can improve π -electron delocalization,^[37,38] the adsorption of Y7-BO with chlorinated end-groups causes more charge transfer between the molecule and the perovskite surface compared to Y6-BO with fluorinated end-groups, leading to larger adsorption energy (Figure 2c). Then, XPS analysis was carried out to further investigate the interaction between NFAs and perovskite, and the results were shown in Figure 2d and 2e. Upon NFAs treatment, both Pb 4f and I 3d peaks shift to lower binding energies, suggesting an increase in the electron density around $[\text{PbI}_6]^{4-}$ octahedron of perovskite from the NFA molecules.^[21] The magnitude of the shift towards lower binding energy follows the order of Y7-BO > Y6-BO, indicating that Y7-BO can bond with perovskite more strongly, in agreement with the DFT results. The Y7-BO modifier exhibiting stronger bonding strength are more favorable to passivate surface defects. As seen in Figure 2f, the result of space-charge-limited current (SCLC) based on electron-only devices with an architecture of FTO/SnO₂/perovskite/PCBM/BCP/Ag (FTO refers to fluorine-doped tin oxide, SnO₂ stands for tin (IV) oxide, and BCP represents bathocuproine) shows that the trap densities are $1.07 \times 10^{14} \text{ cm}^{-3}$, $9.93 \times 10^{13} \text{ cm}^{-3}$ and $9.15 \times 10^{13} \text{ cm}^{-3}$ for the control, Y6-BO-treated and Y7-BO-treated perovskite films,

respectively, and the film thicknesses were obtained from the cross-sectional SEM images in Figure S7. The lower defect density of the PVK/Y7-BO film reveals that the Y7-BO NFAs effectively passivate defects on the perovskite surface. Subsequently, steady-state photoluminescence (PL) and time-resolved PL (TRPL) measurements were performed to directly explore the passivation effect of NFAs. Compared to the control film, PVK/Y6-BO and PVK/Y7-BO films show significantly higher PL intensity (Figure S8) and longer average decay lifetimes (Figure 2g and Table S1), meaning that nonradiative recombination is decreased owing to reduced defects on surface and GBs.^[39] Where the PVK/Y7-BO film exhibits the highest PL intensity and largest lifetime, implying the excellent defect passivation. To quantify the surface nonradiative recombination energy loss, we estimated the quasi-Fermi level splitting (QFLS) by measuring the photoluminescence quantum yield (PLQY) of three type films.^[40] As shown in Figure 2h and 2i, the PVK/Y7-BO films presented the maximum PLQY value of 24.31 % and QFLS value of 1.219 eV, which is expected to increase the open-circuit voltage (V_{oc}) of the resultant photovoltaic device.

Besides optimizing the upper surface of perovskite film, NFAs were also expected to chemical interaction with

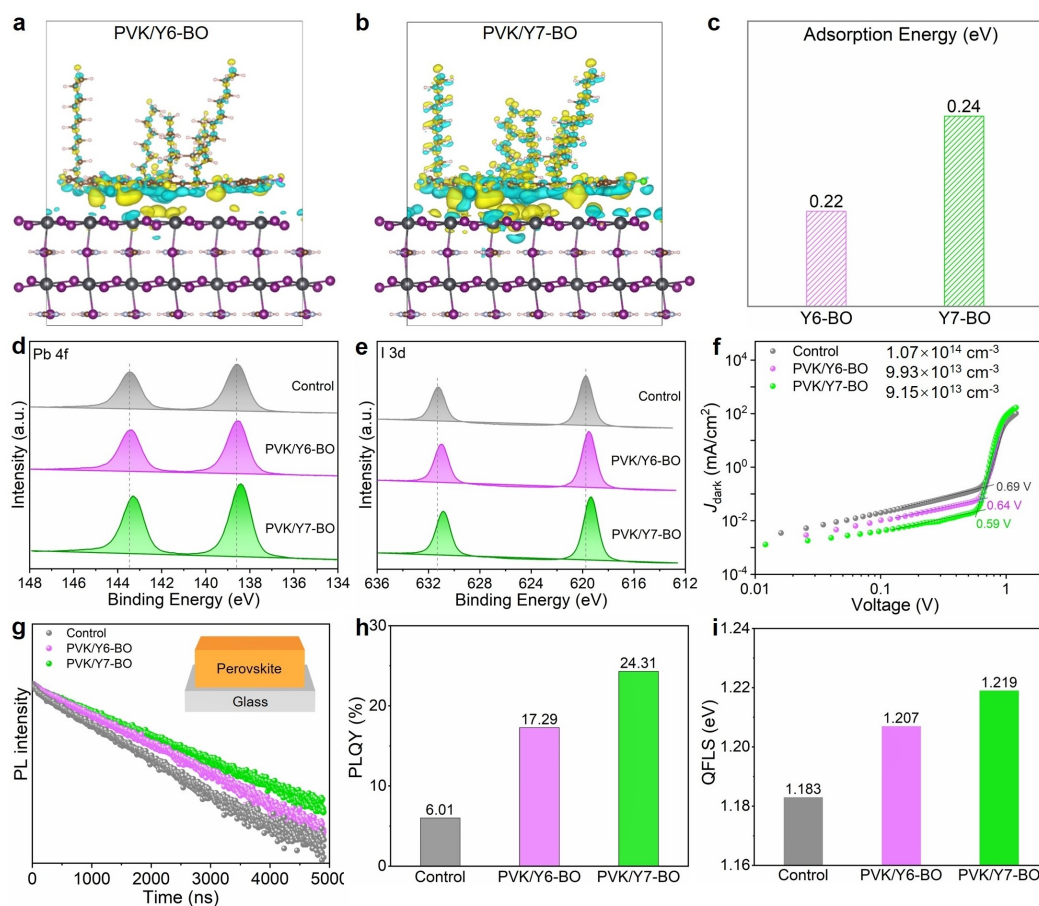


Figure 2. (a-b) Charge density difference (CDD) distributions of Y6-BO and Y7-BO adsorbed on the perovskite surface, respectively. yellow and cyan region represent charge accumulation and depletion. (c) The adsorption energies (E_{ads}) of Y6-BO and Y7-BO adsorbed on the perovskite surface. XPS spectra of (d) Pb 4f and (e) I 3d, (f) Dark J -V curves of the electron-only devices, (g) TRPL spectra, (h) PLQY, and (i) calculated QFLS of PVK, PVK/Y6-BO and PVK/Y7-BO films.

PCBM, affecting the buried surface and thus the morphology of the PCBM films. Figure S9 shows the core-level XPS spectra of O 1s for PCBM films slightly shifted to a lower binding energy compared to the PCBM films with buried surface modified by Y6-BO and Y7-BO (Y6-BO/PCBM and Y7-BO/PCBM films), confirming the coordination between the NFAs and PCBM. Then the effect of NFA modifiers on the morphology of PCBM films was deeply studied by the atomic force microscope (AFM) images, and the results were shown in Figure 3a–c. The PCBM molecules tend to aggregate, and leading to rough surface with a larger root-mean-square (RMS) roughness in the pure PCBM film. While the morphologies of Y6-BO- and Y7-BO-modified PCBM films became smoother. The CB-solvent based PCBM could redissolve the NFAs during the coating process, thus binary NFA:PCBM blend system was formed at the buried surface of PCBM layer, inhibiting the aggregation of PCBM molecules and thus achieving more uniform films. Moreover, PCBM molecule interacting NFAs in the NFA:PCBM blend system, rendering the flexible side chains of NFAs to adjust configuration to fill in the inter-space between the spherical PCBM molecules (Figure 3e–f and Figure S10), leading to higher adsorption energies of

PCBM molecule on the surface of PVK/P6-BO and PVK/Y7-BO layers (Figure 3g). On the contrary, the pure PCBM molecules cannot completely passivate surface defect, due to the weak interactions between PCBM and perovskite (Figure 3d and Figure 3g). The higher adsorption energies suggest that the PVK/NFAs/PCBM interface is theoretically more stable than the PVK/PCBM interface, which is beneficial to inhibit the penetration of water and oxygen, and thus enable the stability of the corresponding photovoltaic devices, combining with the reduced defects on the upper surface of perovskite layer and improved morphology of PCBM layer. In addition, the improved buried surface and morphology of the PCBM films induced by NFA interfacial modifiers can further promote electron transfer from perovskite to PCBM. Subsequently, we employed time resolved PL (TRPL) measurement to study the electron-extraction of these perovskite films with PCBM ETL capping (Figure 3h). The TRPL curves can be fitted by two exponent decays where the fast decay attributed to the charge extraction and slow decay related to radiative recombination (The fitting results were listed in Table S2).^[41] The average lifetime for the PVK sample was 12.17 ns, which decreased to 5.81 and 4.09 ns for PVK/Y6-BO and

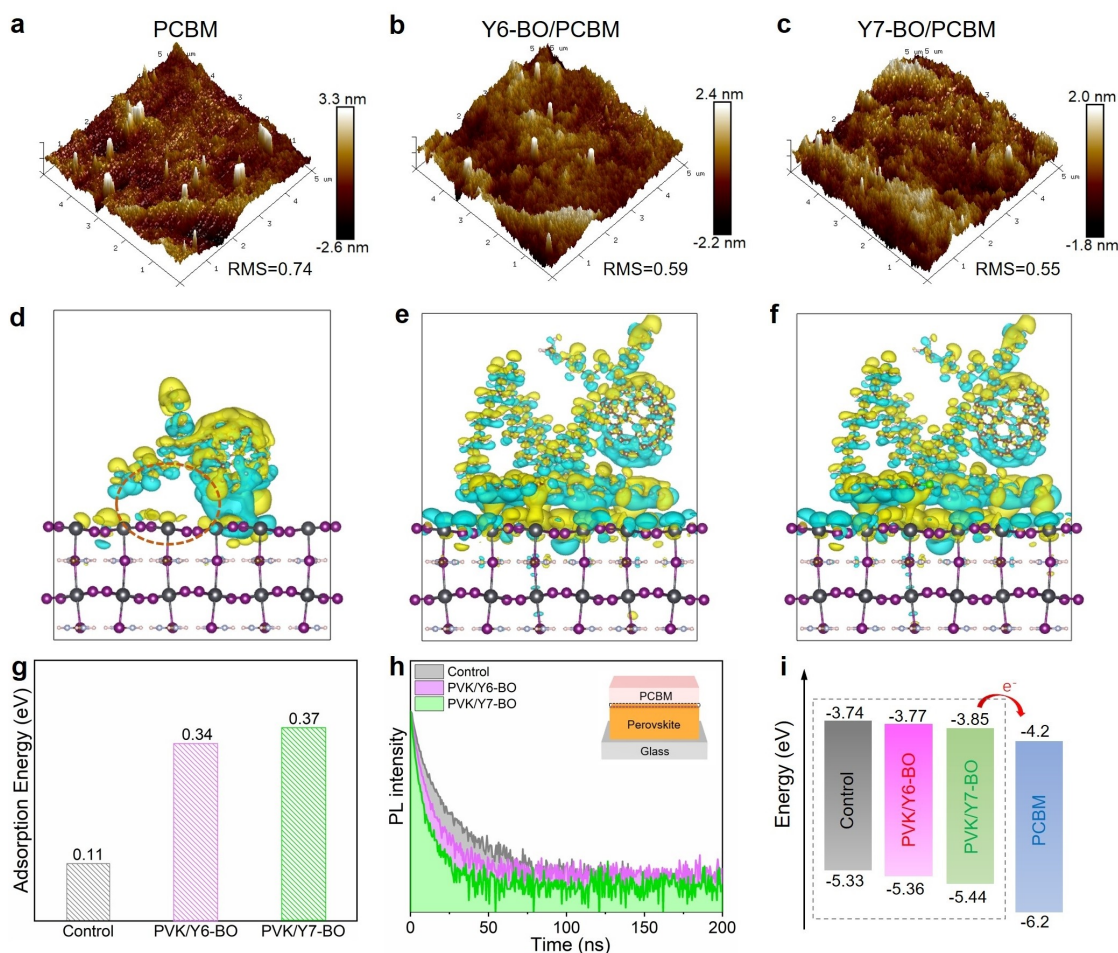


Figure 3. (a–c) AFM images of PCBM, Y6-BO/PCBM, and Y7-BO/PCBM films (the size of images is $5 \times 5 \mu\text{m}$). (d–f) Charge density difference (CDD) distributions of the PCBM covered on the Control, PVK/Y6-BO and PVK/Y7-BO films, respectively. (g) The adsorption energies of PCBM adsorbed on the Control, PVK/Y6-BO and PVK/Y7-BO surface. (h) TRPL spectra of Control, PVK/Y6-BO and PVK/Y7-BO films. (i) Energy-level alignment of Control, PVK/Y6-BO and PVK/Y7-BO films.

PVK/Y7-BO samples, respectively. These results confirm that electron transfer was significantly improved at the perovskite/PCBM interface with NFA modifiers. The Y7-BO can provide best interface properties and appropriate energy level alignment (Figure 3i) obtained from ultraviolet photoelectron spectroscopy (UPS, Figure S11), resulting in the most effective electron extraction.

The effects of optimized upper surface of perovskite and buried surface of PCBM by NFAs on the photovoltaic performance were then evaluated by fabricating i-PSCs with a device structure of FTO/SAM/perovskite/PCBM/BCP/Ag (Figure 4a), in which SAM refers to [2-(3,6-Dimethoxy-9H-carbazol-9-yl) ethyl] phosphonic acid (MeO-2PACz). Notably, the PCE of both NFAs treated devices were improved relative to the control device (Figure S12). Figure 4b shows the best current density-voltage (J - V) curves of control, PVK/Y6-BO and PVK/Y7-BO i-PSCs with an active area of 0.1 cm^2 . The champion PCE of 25.82 % was obtained for the PVK/Y7-BO i-PSC, with a high open-circuit voltage (V_{oc}) of 1.203 V, a short-circuit current density (J_{sc}) of 25.62 mA/cm^2 , and a fill factor (FF) of 83.88 %. The PVK/Y6-BO

device achieved a PCE of 25.12 % (with a V_{oc} of 1.185 V, a J_{sc} of 25.12 mA/cm^2 and a FF of 83.09 %). In comparison, the control device achieved a PCE of 24.49 % (V_{oc} : 1.172 V, J_{sc} : 25.50 mA/cm^2 , FF: 81.92 %). Then, the external quantum efficiency (EQE) spectra was used to compare with the J_{sc} from the J - V curves, as shown in Figure 4c. The control i-PSC gave an integrated J_{sc} of 24.23 mA/cm^2 with a larger discrepancy (4.91 %). In contrast, the PVK/Y6-BO and PVK/Y7-BO devices presented a reasonable discrepancy of 3.36 % and 2.88 %, respectively. Notably, the PCE achieved here is the highest value among reported i-PSCs with surface treated by NFAs, as summarized in Figure 4d and Table S3. The increases of J_{sc} and V_{oc} ascribed to the more efficient electron extraction and nonradiative recombination suppression mainly contribute to the PCE improvements of NFA-treated i-PSCs. Whereafter, charge dynamics of the devices were systematically studied. Intensity modulated photo-current spectrum (IMPS) was performed to investigate the charge transfer characteristics of these i-PSCs,^[42] and the results were shown in Figure 4e. The charge transfer time (τ_{tr}) for control, PVK/Y6-BO and PVK/Y7-BO devices were

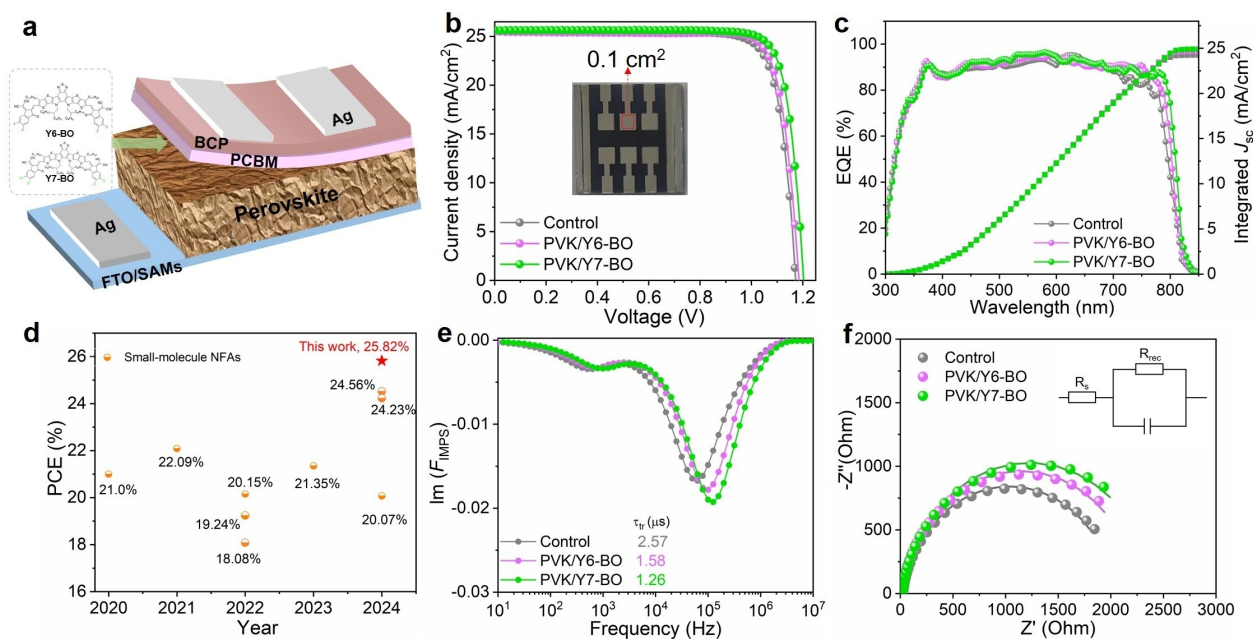


Figure 4. (a) Schematic illustration showing the device architecture. (b) J - V curves and (c) EQE spectra of the champion Control, PVK/Y6-BO and PVK/Y7-BO PSC devices. (d) Summary of PCEs for NFAs surface treated i-PSCs reported in the literatures. (e) IMPS measurements, and (f) Nyquist plots (inset: equivalent circuit) of the Control, PVK/Y6-BO and PVK/Y7-BO devices.

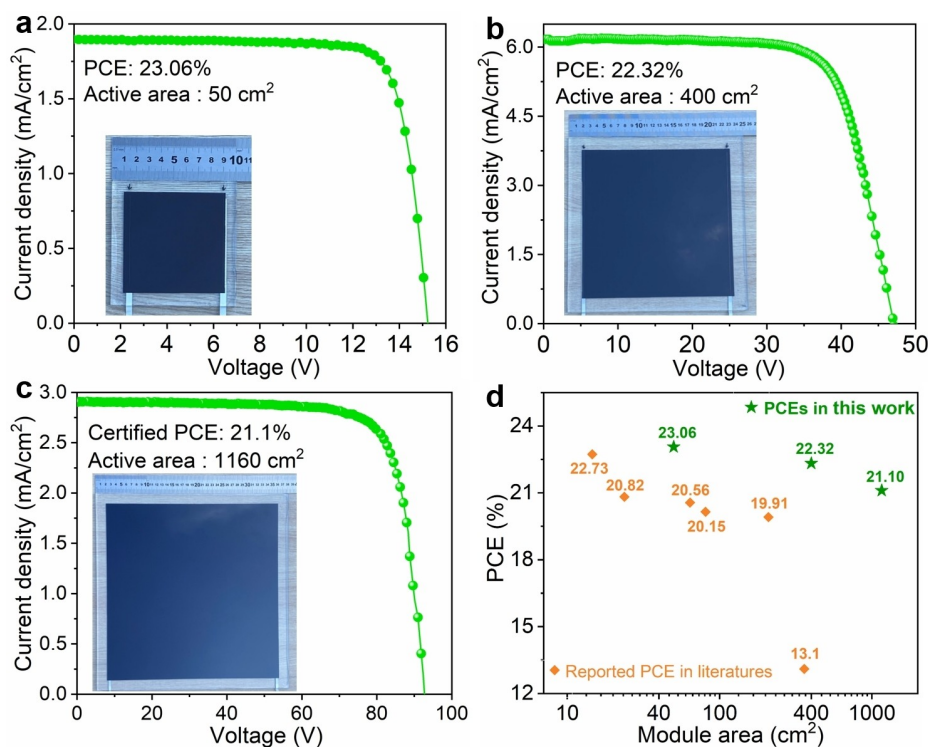


Figure 5. (a–c) J - V characteristics of i-PSMs with different areas of 50 cm², 400 cm², and 1160 cm², respectively. Inset: PCEs and photographs of the device. (d) Summary of reported PCEs for i-PSMs with different areas.

2.57, 1.58, and 1.26 μ s, respectively. The shortest τ_{tr} of PVK/Y7-BO device indicates that the most efficient electron extraction/transport occurred in this i-PSC.^[43] Furthermore, charge recombination property was analyzed by the electro-

chemical impedance spectroscopy (EIS) measurement, and the Nyquist plots of three type i-PSCs were shown in Figure 4f. The inset in Figure 5f is the equivalent circuit to fit the curves, and the fitted results are listed in Table S4.

Herein, charge recombination resistance (R_{rec}) is associated with the charge recombination at interface.^[44] The PVK/Y6-BO and PVK/Y7-BO i-PSCs exhibit smaller R_{rec} values, implying suppressed charge nonradiative recombination in comparison to control device. Accordingly, the suppression of such recombination losses results in the PCE improvement of NFA-based i-PSCs. Obviously, the device with Y7-BO NFA performs the most effective electron extraction and recombination suppression compared with those of control and Y6-BO-based devices, supporting its outstanding efficiency.

The Y7-BO NFA enables the best performances of small-size i-PSCs, therefore, large-area i-PSMs treated by Y7-BO with different areas of 50 cm², 400 cm², and 1160 cm² were prepared, and their photographs were displayed in Figure S13. Figure 5a–5c show the J – V characteristics of the best-performing i-PSMs, their corresponding photovoltaic

parameters are reported in Table 1. The champion PCE of different area i-PSMs are 23.06 %, 22.32 %, and 21.1 %, respectively. Notably, the PCE of 21.1 % for 1160 cm² device was certified by the third party (Figure S14–S15, Supporting Information). To the best of our knowledge, these i-PSMs possess the highest PCE values reported among perovskite solar modules in the literatures (Figure 5d and Table S5).

In addition, we investigated the effects of NFAs on the stability of photovoltaic devices. The air stability of unencapsulated i-PSCs was greatly improved for the device with NFAs (Figure 6a). The Y6-BO and Y7-BO based devices retain their initial PCE over 90 % after aging at the humidity of 40 % for 1400 h, whereas control one retained only 64 % of the initial PCE. Water contact angle in Figure 6c–d demonstrated that the hydrophobicity of perovskite film with NFAs (Y6-BO: 84.6°, Y7-BO: 87.3°) surpassed that of control one (71.3°), contributing to the air stability of the corresponding devices. Moreover, the device's operational stability of i-PSMs at maximum power point (MPP) under a continuous illumination was characterized and shown in Figure 6b. The device with NFAs are more stable compared to the control device. Especially, the Y7-BO based i-PSM showed only 5.6 % efficiency drop after 1522 h. ion migration is the main cause for device degradation,^[27] thus the ion distribution of I[−] element was probed by time-of-flight secondary ion mass spectrometry (TOF-SIMS), as shown in Figure 6f–6h. Strikingly, less I[−] elements distribute in the bulk of perovskite layer and a I[−]

Table 1: Photovoltaic parameters of control, PVK/Y6-BO and PVK/Y7-BO perovskite solar cells, and PVK/Y7-BO perovskite solar modules.

Device type	Area (cm ²)	Perovskite	V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)
Cell	0.1	Control	1.172	25.48	81.92	24.49
		PVK/Y6-BO	1.185	25.52	83.09	25.12
		PVK/Y7-BO	1.203	25.62	83.75	25.82
Module	50		15.223	1.90	79.89	23.06
	400	PVK/Y7-BO	47.124	6.17	76.80	22.32
	1160		92.391	2.91	78.40	21.10

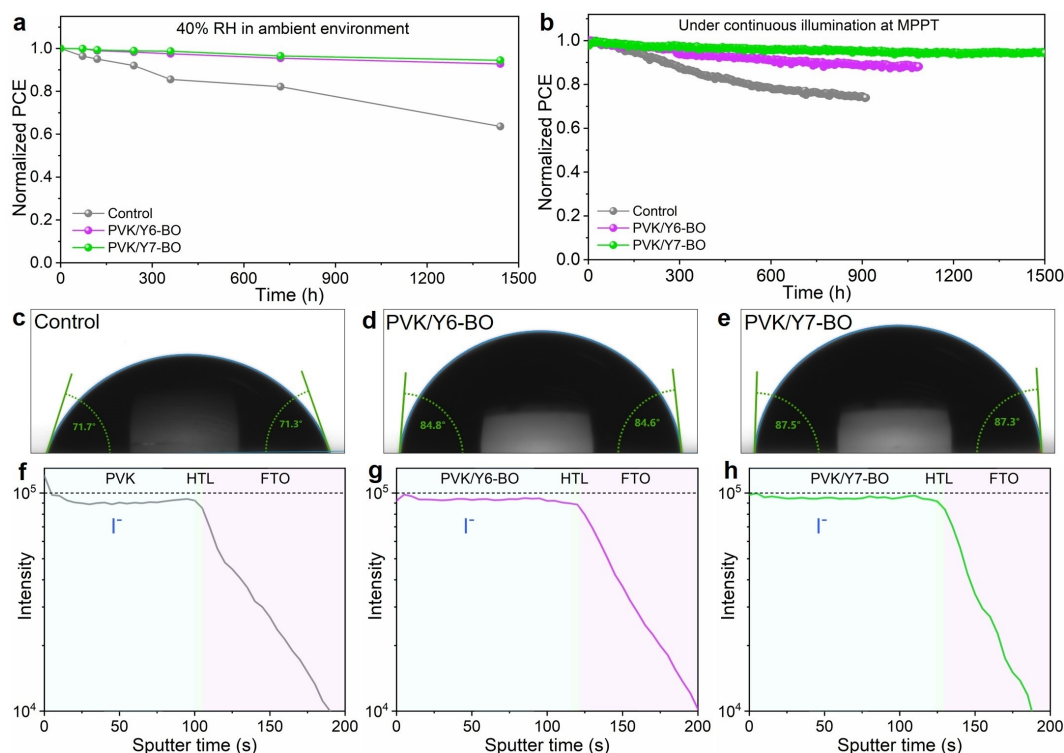


Figure 6. (a) The device air stability of i-PSCs without encapsulation under a relative humidity of ~40% in ambient environment. (b) Operation stability tests of the unencapsulated i-PSMs under continuous light illumination in an N₂ atmosphere at room temperature. (c–e) The surface contact angle measurements of water for of control, PVK/Y6-BO and PVK/Y7-BO films. (f–h) TOF-SIMS results of the control, PVK/Y6-BO and PVK/Y7-BO films.

peak signal is seen on the upper surface of control film, whereas no I^- accumulations are found on the surface of PVK/Y6-BO and PVK/Y6-BO layers, suggesting that the I^- migration may disrupt the device's operational stability. The above support the use of CB solvent system instead of IPA for surface treatment, leading to devices with greater stability. For control devices, the use of a protic polar solvent IPA during surface treatment induces surface reconstruction of the underlying perovskite layer (Figure S16), resulting in a destabilized perovskite/PCBM interface and an undesirable degradation of device properties. On the contrary, the employing NFAs modifiers in CB solvent ensure that the underlying perovskite layer remains pristine during surface treatment without negatively affecting, which also enhance the connection between perovskite and PCBM and improve the morphology of PCBM. Therefore, the stabilized perovskite/PCBM interface by Y6-BO and Y7-BO modifiers can efficiently suppress moisture infiltration and I^- migration, preventing the perovskite degradation and boosting the long-term device stability.

Conclusion

In summary, Y6-BO and Y7-BO NFAs were introduced as interfacial modifiers between perovskite and PCBM layers for improving the photovoltaic efficiency and stability in i-PPVs. The NFAs in non-polar CB solvent can provide effective defects passivation without disrupting the underlying perovskite film and simultaneously optimize the buried surface of PCBM layer, significantly promoting electron extraction at the top interface and thus improving device performance. More importantly, Y7-BO NFBA can extend the interaction between perovskite and PCBM layers, benefiting device performances. As a result, the Y7-BO-based i-PSC achieves a record PCE of 25.82% among reported i-PSCs with interface modified by NFAs. Remarkably, outstanding efficiencies of 23.06%, 22.32%, and 21.10% were obtained for the Y7-BO-based i-PSMs with active areas of 50 cm², 400 cm², and 1160 cm², respectively. Moreover, the ambient and operational stabilities of i-PPVs with NFAs were significantly improved, thanks to the interfacial passivation without disrupting the underlying perovskite layer, improved uniform of PCBM layer, and enhanced connection between perovskite and PCBM layers. This work provide an promising prospect for the scaled-up production of perovskite solar cells to modules.

Supporting Information

Supporting Information is available from Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: perovskite solar cells and modules • device stability • interface modification • electron transport layer • non-fullerene acceptor

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